Transformer Model

Formula sheet for a three-phase two-winding transformer

Notation. Indices $k = 1, 2, 3$ identify the transformer legs or phases; p and s denote the primary and secondary windings. Uppercase U and I denote electrical quantities, Φ the magnetic flux, $\Psi = N\Phi$ the flux linkage, and $\Theta = NI$ the magnetomotive force. The example uses vector group Yy0.

Transformer-leg equations

For each leg $k \in \{1, 2, 3\}$, the reluctance associated with the iron-loss resistance is

$$\mathcal{R}_{R\mathrm{Fe}} = \frac{N_1^2}{R_{\mathrm{Fe}}}. \tag{1}$$

The nonlinear magnetizing characteristic is represented by

$$\Theta_{\mu k} = N_1 I_{\mu N} \left[a \frac{\Phi_k}{\Phi_N} + (1 - a) \left(\frac{\Phi_k}{\Phi_N} \right)^m \right], \qquad k = 1, 2, 3. \tag{2}$$

Equivalently, the magnetizing current of one leg is

$$I_{\mu k} = I_{\mu N} \left[a \frac{\Phi_k}{\Phi_N} + (1 - a) \left(\frac{\Phi_k}{\Phi_N} \right)^m \right]. \tag{3}$$

Primary-winding equations

For every phase k,

$$\Phi_{kp} = \frac{\Psi_{kp}}{N_1} = \frac{1}{N_1} \int (U_{kp} - R_1 I_{kp}) \, \mathrm{d}t, \qquad I_{kp} = \frac{\Theta_k}{N_1}. \tag{4}$$

Written separately for the three phases, this relation applies for $k = 1$, 2, and 3 with the corresponding phase quantities.

Secondary-winding equations

For every phase k,

$$\Phi_{ks} = \frac{\Psi_{ks}}{N_2} = \frac{1}{N_2} \int (U_{ks} - R_2 I_{ks}) \, \mathrm{d}t, \qquad I_{ks} = \frac{\Theta_k}{N_2}. \tag{5}$$

The voltage and current reference directions must be chosen consistently with the winding polarities used in the circuit model.

Common core flux and winding flux linkages

For an ideal magnetic coupling of the windings on each leg,

$$\Phi_{kp} = \Phi_{ks} = \Phi_k, \qquad k = 1, 2, 3. \tag{6}$$

The corresponding flux linkages are

$$\Psi_{kp} = N_1 \Phi_k, \qquad \Psi_{ks} = N_2 \Phi_k. \tag{7}$$

Thus, in general,

$$\Psi_{kp} \neq \Psi_{ks} \quad \text{when} \quad N_1 \neq N_2, \tag{8}$$

even though both windings link the same core flux Φ_k.

Leakage-flux branch

With $\mathcal{R}_{\Phi\sigma}$ denoting the equivalent magnetic reluctance of the leakage path,

$$\Phi_{k\sigma} = \frac{\Theta_k}{\mathcal{R}_{\Phi\sigma}}, \qquad k = 1, 2, 3. \tag{9}$$

For a leakage inductance $L_{1\sigma}$ referred to the primary winding,

$$L_{1\sigma} = \frac{N_1^2}{\mathcal{R}_{\Phi\sigma}} \qquad \Longleftrightarrow \qquad \mathcal{R}_{\Phi\sigma} = \frac{N_1^2}{L_{1\sigma}}. \tag{10}$$

Tank representation

The tank is represented by an additional magnetic reluctance $\mathcal{R}_{\text{tank}}$. The source text recommends the following approximate choices:

$$\mathcal{R}_{\text{tank}} \approx (0.1 \ldots 0.01)\frac{N_1^2}{L_{1\sigma}}, \qquad \text{rough initial estimate - (10x-100x leakage inductance),} \tag{11}$$

$$\mathcal{R}_{\text{tank}} \approx 10^5 \ldots 10^6, \qquad \text{tank magnetic reluctance in the model units,} \tag{12}$$

$$\mathcal{R}_{\text{tank}} \approx 10^{20}, \qquad \text{tank branch absent (practically open).} \tag{13}$$

Rated data: 300 MVA, 400 kV / 120 kV, 50 Hz

Symbol	Value	Meaning
S_N	300×10^6 VA	Three-phase rated apparent power
$U_{1N,LL}$	400×10^3 V	Primary rated line-to-line voltage
$U_{2N,LL}$	120×10^3 V	Secondary rated line-to-line voltage
f_N	50 Hz	Rated frequency
N_1	1000	Primary number of turns
u_R	0.002	Resistive part of relative short-circuit voltage
u_X	0.2	Inductive part of relative short-circuit voltage
i_0	0.0006	Relative no-load current
p_0	0.0004	Relative no-load loss, with $p_0 < i_0$
a	0.75	Linear weighting in the magnetizing characteristic
m	9	Exponent of the nonlinear magnetizing term

Per-leg quantities for vector group Yy0

The three-phase values are first converted to one transformer leg:

$$S_{N,\mathrm{leg}} = \frac{S_N}{3}, \qquad U_{1N} = \frac{U_{1N,LL}}{\sqrt{3}}, \qquad U_{2N} = \frac{U_{2N,LL}}{\sqrt{3}}, \tag{14}$$

$$I_{1N} = \frac{S_{N,\mathrm{leg}}}{U_{1N}}, \qquad Z_{1N} = \frac{U_{1N}}{I_{1N}}, \qquad \omega_N = 2\pi f_N. \tag{15}$$

Winding and iron-loss parameters

The short-circuit impedance is split equally between the two windings and referred initially to the primary side:

$$R_1 = \frac{Z_{1N} u_R}{2}, \qquad L_{1\sigma} = \frac{Z_{1N} u_X}{2\omega_N}, \tag{16}$$

$$R_{\mathrm{Fe}} = \frac{U_{1N}^2}{p_0 S_{N,\mathrm{leg}}}, \qquad \mathcal{R}_{R\mathrm{Fe}} = \frac{N_1^2}{R_{\mathrm{Fe}}}, \tag{17}$$

$$N_2 = N_1 \frac{U_{2N}}{U_{1N}}, \qquad R_2 = R_1 \left(\frac{N_2}{N_1}\right)^2, \tag{18}$$

$$\mathcal{R}_{\Phi\sigma} = \frac{N_1^2}{L_{1\sigma}}. \tag{19}$$

Magnetizing branch and rated flux

From the no-load current and no-load losses,

$$X_h = \frac{U_{1N}^2}{S_{N,\mathrm{leg}}\sqrt{i_0^2 - p_0^2}}, \tag{20}$$

$$I_\mu = \frac{U_{1N}}{X_h}, \tag{21}$$

$$I_{\mu N} = \sqrt{2}\, I_\mu, \tag{22}$$

$$\Phi_N = \frac{\sqrt{2}\, U_{1N}}{\omega_N N_1}. \tag{23}$$

The nonlinear characteristic then becomes

$$I_\mu(\Phi) = I_{\mu N}\left[a\frac{\Phi}{\Phi_N} + (1-a)\left(\frac{\Phi}{\Phi_N}\right)^m\right]. \tag{24}$$

Parameter definitions

Symbol	Meaning
Θ_k	Magnetomotive force of leg k
Φ_k, $\Phi_{k\sigma}$	Main and leakage fluxes of leg k
Ψ_{kp}, Ψ_{ks}	Primary and secondary flux linkages
$\mathcal{R}_{R\mathrm{Fe}}$	Magnetic reluctance associated with the iron-loss branch
$\mathcal{R}_{\Phi\sigma}$	Equivalent leakage-path reluctance
$\mathcal{R}_{\mathrm{tank}}$	Tank-path magnetic reluctance
R_1, R_2	Primary and secondary winding resistances
R_{Fe}	Equivalent iron-loss resistance
$L_{1\sigma}$	Primary-referred leakage inductance
X_h	Main magnetizing reactance
$I_{\mu N}$, Φ_N	Rated peak magnetizing current and rated peak flux
a, m	Shape parameters of the nonlinear magnetization curve

Implementation notes. The time integrals may be implemented with behavioural sources in LTspice. When forming the winding equations, verify the chosen voltage polarities, current reference directions, initial flux, and the sign of each magnetomotive-force contribution. The numerical tank-reluctance range is model- and unit-dependent and should be validated against the intended zero-sequence behaviour.

Typeset from the supplied transformer-model notes. English wording and notation have been standardized; explicitly identified dimensional inconsistencies in the source have been corrected.